IBM Computer Usability Satisfaction Questionnaires: Psychometric Evaluation and Instructions for Use

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This article describes recent research in subjective usability measurement at IBM, focused on evaluating the psychometric properties of questionnaires designed for use in scenario-based usability evaluation. The questionnaires address evaluation at both a global overall system level and at a more detailed scenario level. The primary goals of this article are to (a) discuss the psychometric characteristics of IBM questionnaires that measure user satisfaction with computer system usability, and (b) provide the questionnaires, with administration and scoring instructions. For scenario-level measurement, the 3-item After-Scenario Questionnaire (ASQ) has excellent internal consistency, with coefficient alphas across a set of scenarios ranging from .90 to .96. For more global assessment, the Post-Study System Usability Questionnaire (PSSUQ) also has excellent internal consistency, with an overall coefficient alpha of .97. Preliminary principal factor analysis of 48 PSSUQ questionnaires suggested the presence of three fac-

This article brings together work that many human factors engineers and other usability professionals (both inside and outside IBM) reviewed for publication as technical reports and conference papers. These reviews substantially improved the presentation of this research. I do want to acknowledge, specifically, the contributions of several key individuals. First, this research would never have taken place without the management support of Robert Mack at the T. J. Watson Research Center's User Interface Institute. While I was on assignment at the research center, Dr. Mack also gave me the opportunity to study multivariate analysis (with Jane Monroe) and psychometric methods (with Marvin Sontag) at the Teacher's College, Columbia University, Department of Measurement and Statistics. After I returned to Boca Raton, Don Davis, then manager of the Design Center, Human Factors Department, gave me the flexibility to continue analyzing the office-applications data collected at the T. J. Watson Research Center. Suzanne Henry made her office-applications ASQ and PSSUQ data available to me, providing a database large enough for a preliminary psychometric evaluation. Richard Granda, Suzanne Henry, Mary LaLomia, Alan Happ, and Sharon Stanners all contributed to the effort to name the PSSUQ factors. I am especially grateful to Dr. LaLomia for her encouragement to pursue this work and for her valuable technical comments, and for the comments of the reviewers for IJHCI who worked their way through several versions of this article.

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tors named, after varimax rotation, System Usefulness, Information Quality, and Interface Quality, with corresponding coefficient alphas of .96, .91, and .91. Evaluation of 377 PSSUQ questionnaires (modified to allow mailing to respondents in their offices and referred to as the Computer System Usability Questionnaire, or CSUQ) confirmed the structure of the preliminary principal factor analysis. Consequently, usability practitioners can use these questionnaires to help them measure users' satisfaction with the usability of computer systems in the context of scenario-based usability studies.

INTRODUCTION

Customers want usable products, and developers strive to produce them. It follows that an important part of modern product engineering, both hardware and software, must be the measurement of usability. Measuring usability is particularly difficult because usability is not a unidimensional product or user characteristic, but emerges as a multidimensional characteristic in the context of users performing tasks with a product in a specific environment (Bevan, Kirakowski, & Maissel, 1991; Shackel, 1984). However, if you are unable to measure usability, how can you judge your product against your competitors', or even your own previous versions of the product? The appropriate measurement methods for assessing usability are not obvious, and are an ongoing concern of human factors engineers involved in the development of computer systems.

A common approach to usability assessment is to ask a set of participants to complete a set of usability assessment scenarios using the system of interest. Usability assessment scenarios are sets of tasks in which users solve realistic problems, such as how to create and print a document or how to compose and send a piece of electronic mail. Researchers can measure usability in the context of a scenario-based evaluation with both objective and subjective variables. Objective usability measures include, but are not limited to, scenario completion time, successful scenario completion rate, and time spent recovering from errors (Whiteside, Bennett, & Holtzblatt, 1988). Subjective usability measures are usually responses to Likert-type questionnaire items that assess user attitude concerning attributes such as system ease of use and interface likeability (Alty, 1992). Most usability evaluators collect both objective and subjective data. Which type of measure is "best" depends on the purpose of the evaluation. If the development goal of a new system is to increase productivity, then objective measures are of primary importance. If the development goal is user satisfaction, then subjective measures are more important. In either case, it is important to understand the psychometric properties of the measures.

The goal of psychometrics is to establish the quality of psychological measures (Nunnally, 1978). Is a measure reliable in the sense that it is consistent? Given a reliable measure, is it valid (measures the intended attribute)? Finally, is the measure appropriately sensitive to experimental manipulations? Psychomet-

rics is a well-developed field, but usability researchers have only recently used these methods to develop and evaluate questionnaires to assess usability (Sweeney & Dillon, 1987).

Brief Review of Psychometric Practice

Reliability Goals. In psychometrics, a questionnaire's reliability is a quantitative assessment of its consistency. After determining the relationship between items and factors (with factor analysis, discussed later), the items that make up a factor are the component items of a corresponding summative scale. The most common way to estimate the reliability of these types of scales is with coefficient alpha (Nunnally, 1978). Coefficient alpha can range from 0 (no reliability) to 1 (perfect reliability). Measures that can affect an individual's future, such as IQ tests or college entrance exams should have a minimum reliability of .90, and preferably, a reliability of .95. For other research or evaluation, measurement reliability should be .70 to .80 (Landauer, 1988; Nunnally, 1978).

Validity Goals. A questionnaire's validity is the extent to which it measures what it claims to measure. Researchers commonly use the Pearson correlation coefficient to assess criterion-related validity (the relationship between the measure of interest and a different concurrent or predictive measure). This correlation does not have to be large to provide evidence of validity. For personnel selection decisions—decisions that may have a serious effect on an individual's life—moderate correlations (with absolute values as small as .30 to .40) are large enough to justify the use of psychometric instruments (such as questionnaires; Nunnally, 1978). A validity coefficient is concurrent if the measurements occur at the same time, and is predictive if the measure of interest precedes the criterion measurement. An instrument can be reliable without being valid, but cannot be valid unless it is reliable. The statistical reliability of an instrument and the reliability of the criterion measurement set the upper limit of the criterion-related validity between the measures. This is one reason that criterion-related validity coefficients tend to appear low (Nunnally, 1978).

Number of Scale Steps. All other things being equal, the more scale steps the better, but with rapidly diminishing returns. Numerous studies show that the reliability of individual rating scales is a monotonically increasing function of the number of steps (Nunnally, 1978). As the number of scale steps increases from 2 to 20, the increase in reliability is very rapid at first, but tends to level off at about 7. After 11 steps there is little gain in reliability from increasing the number of steps. The question of the number of steps on a rating scale is very important if there is only one scale, but is usually less important when summing scores over a number of scales. Attitude scales tend to be highly reliable because the items tend to correlate rather highly with one another. Reliability, then, usually is not a serious problem in the construction of summated attitude scales (Nunnally, 1978). Many re-

searchers use 7 scale steps as the appropriate balance between scale reliability and discriminative demand on the respondent.

Factor Analysis. Factor analysis is a statistical procedure that examines the correlations among variables to discover clusters of related variables (Nunnally, 1978). Because summated (Likert) scales are more reliable than single-item scales (Nunnally, 1978), and it is easier to present and interpret a smaller number of scores, it is common to conduct a factor analysis to determine if there is a statistical basis for the formation of summative scales. Generally, a factor analysis requires at least 5 participants per item to ensure stable factor estimates (Nunnally, 1978).

One way to determine the appropriate number of factors in a factor analysis is to graph a scree plot of the factors' eigenvalues (variances; Cliff, 1987; Nunnally, 1978). The point at which the slope of the line becomes discontinuous indicates an appropriate number of factors for the analysis (Cliff, 1987). Discontinuity analysis is one of the more effective means for determining the number of factors (Coovert & McNelis, 1988). The basic approach of discontinuity analysis is to calculate the differences between adjacent eigenvalues. A discontinuity occurs when the magnitude of a difference exceeds the immediately preceding difference. When this happens, it is reasonable to retain the same number of factors as eigenvalues that precede the discontinuity. Researchers should avoid using the common rule-of-thumb (and common computer package default) to let the number of factors equal the number of eigenvalues that are greater than 1 (Cliff, 1987; Coovert & McNelis, 1988).

Calculating Scale Scores. From psychometric theory (Nunnally, 1978), scale reliability is a function of the interrelatedness of scale items, the number of scale steps per item, and the number of items in a scale. If a participant chooses not to answer an item, the effect would be to reduce slightly the reliability of the scale in that instance. In most cases, the remaining items should offer a reasonable estimate of the appropriate scale score. From a practical standpoint, averaging the answered items to obtain the scale score enhances the flexibility of use of the questionnaire, because if an item is not appropriate in a specific context and users choose not to answer it, the questionnaire is still useful. Also, users who do not answer every item can stay in the sample. Finally, averaging items to obtain scale scores does not affect the statistical properties of the scores, and standardizes the range of scale scores. For example, with items based on 7-point scales, all the summative scales would also have scores that range from 1 to 7, regardless of the number of items per scale. This standardization makes scale scores easier to interpret and compare.

Current Psychometric Instruments for Usability Evaluation of Computer Systems

LaLomia and Sidowski (1990) reviewed six instruments that measure computer satisfaction. The Gallagher Value of MIS Reports Scale and the Larcker and Lessig

Perceived Importance and Usableness of MIS Reports Scales address satisfaction with management information system (MIS) reports, and are clearly not applicable to most scenario-based usability evaluations. The Hatcher and Diebert Computer-Acceptance Scale measures how office personnel accept using a computer. The Pearson and Bailey User Satisfaction Scale is a comprehensive measure, but was developed to assess user satisfaction specifically with mainframe computer systems, and requires about 20 min to complete.

Chin, Diehl, and Norman (1988) evaluated the Questionnaire for User Interface Satisfaction (QUIS) Version 5.0. The QUIS is a 27-item questionnaire using 10-point bipolar scales, organized into five sections (Overall Reaction, Screen, Terminology and System Information, Learning, and System Capabilities). Principal factor analysis of the QUIS items for Screen, Terminology and System Information, Learning, and System Capabilities (21 items, 96 participants) indicated the presence of four factors, roughly corresponding to these four sections of the questionnaire. One screen item and one terminology item failed to load on any factor. One system capabilities item loaded with the learning items. One system capabilities item, one terminology item, and one screen item comprised the third factor. Overall, the internal consistency of the QUIS was .94. No information is available regarding subscale reliability (basing subscales on either the questionnaire sections or factors). The lack of correspondence between item groups and underlying factors and the paucity of information regarding subscale reliability are problems that limit the usefulness of the QUIS.

The Human Factors Research Group (HFRG) at University College Cork (Kirakowski & Dillon, 1988) developed the Computer Satisfaction Inventory (CUSI). The CUSI is a 22-item questionnaire containing two subscales, Affect and Competence (determined using cluster analysis on ratings from 195 respondents). The overall internal consistency was .94, with .91 for Affect and .89 for Competence. Unlike the QUIS, researchers who wish to use the CUSI must purchase it from the HFRG, which limits its general availability and usefulness.

Current Research Focus

The focus of this research was the application of psychometric methods to the development and evaluation of standard questionnaires to assess subjective usability. In contrast to other recent computer-user satisfaction questionnaires (Chin et al., 1988; Kirakowski & Dillon, 1988) the IBM questionnaires are specifically for use in the context of scenario-based usability evaluation (Lewis, 1991a, 1991b, 1992b; Lewis, Henry, & Mack, 1990), although additional research has indicated that one may be useful as an instrument for field evaluation (Lewis, 1992a). The goal of this article is to make these questionnaires and their psychometric properties generally available to usability practitioners. (The questionnaires appear in the Appendix.)

THE AFTER-SCENARIO QUESTIONNAIRE (ASQ)

In a scenario-based usability study, participants use a product, such as a computer application, to do a series of realistic tasks. The After-Scenario Questionnaire (ASQ) is a 3-item questionnaire that IBM usability evaluators have used to assess participant satisfaction after the completion of each scenario (see Appendix for a copy of the questionnaire). The items address three important aspects of user satisfaction with system usability: ease of task completion, time to complete a task, and adequacy of support information (on-line help, messages, and documentation). Because the questionnaire is very short, it takes very little time for participants to complete, an important practical consideration for usability studies. Usability professionals have used these items (or very similar items) in usability studies at IBM for many years, but a recent series of studies has provided a database of sufficient size to allow a preliminary psychometric evaluation of the ASQ.

The ASQ items are the constituent items for a summative (or Likert-type) scale (McIver & Carmines, 1981; Nunnally, 1978). In developing summative scales, it is important to consider item construction, item selection, and psychometric evaluation.

Item Construction

The items are 7-point graphic scales, anchored at the end points with *strongly agree* (1) and *strongly disagree* (7), and a *not applicable* (N/A) point outside the scale, as shown in the Appendix.

Item Selection

The content of the items reflects aspects of usability that usability professionals at IBM have generally considered important.

Psychometric Evaluation

The Office-Applications Studies. Scenario-based usability studies of three office application systems (Lewis et al., 1990) provided the data for a psychometric evaluation of the ASQ. Forty-eight employees of temporary help agencies participated in the studies, with 15 hired in Hawthorne, New York, 15 hired in Boca Raton, Florida, and 18 hired in Southbury, Connecticut. Each set of participants consisted of one-third clerical/secretarial work experience with no mouse experience (SECNO), one-third business professionals with no mouse experience (BPNO), and one-third business professionals with at least 3 months mouse experience (BPMS). All participants had at least 3 months experience using some type

of computer system. They had no programming training or experience, and had no (or very limited) knowledge of operating systems.

Popular word-processing applications, mail applications, calendar applications, and spreadsheet applications installed in three different operating environments comprised the three office systems (hereafter referred to as System I, System II, and System III). The participants in Hawthorne used System I, those in Boca Raton used System II, and those in Southbury used System III, resulting in a confounding of location and system. (This confounding potentially affects conclusions drawn regarding the systems, but has no effect on the psychometric evaluation of the questionnaires.)

All three systems allowed windowing, used a mouse as a pointing device, and allowed a certain amount of integration among the applications. The systems differed in details of implementation, but were generally similar. The three word-processing and spreadsheet applications were similar, but the mail and calendar applications differed considerably. The studies contained eight scenarios in common, listed in Table 1.

Participants began the study with a brief lab tour, read a description of the study's purpose and the day's agenda, and completed a background questionnaire. Participants using System I completed an interactive tutorial shipped with the system. Tutors provided the other participants with a brief demonstration about how to move, point, and select with a mouse, how to open the icons for each product, and how to maximize and minimize windows. After this system exploration period (usually about 1 hr), participants performed the scenarios, completing the ASQ as they finished each scenario. While the participant performed the scenario, an observer logged the participant's activities. If the participant completed the scenario without assistance and produced the correct output, then he or she completed the scenario successfully. Either after completing all scenarios or at the end of the workday (with some scenarios never attempted), participants provided an overall system rating with the Post-Study System Usability Questionnaire (PSSUQ; Lewis, 1992b; Lewis et al., 1990). Participants usually needed a full work day (8 hr) to complete the study.

Table 1. Descriptions of the Eight Office-Applications Scenarios

Scenario Type	Component Tasks
Mail (M1A)	Open, reply to, and delete a note.
Mail (M1B)	Open, reply to, and delete a note.
Mail (M2)	Open a note, forward with reply, save and print the note.
Address (A1)	Create, change, and delete address entries.
File Management (F1)	Rename, copy, and delete a file.
Editor (E1)	Create and save a short document.
Editor (E2)	Locate and edit a document, open a note, copy text from the note into the document, save and print the document.
Decision Support (D1)	Create a small spreadsheet, open a document, copy the spreadsheet into the document, save and print the document, save the spreadsheet

At the end of the three studies, the researchers entered the responses to the ASQ, PSSUQ, and the scenario completion data into a database. From this database, it was possible to analyze the ASQ's reliability, validity, and sensitivity.

Reliability. It makes little sense to attempt to derive factors for a 3-item scale, but it is possible to assess the scale's reliability using coefficient alpha, calculating coefficient alpha for each of the eight scenarios. The eight coefficient alphas ranged from .90 to .96 and averaged .93. Coefficient alphas this large were surprising because each scale contained only three items, and reliability is largely a function of the number of scale items (Nunnally, 1978). High coefficient alphas with few items indicate high interitem correlations, a desirable trait in a psychometric instrument.

Validity. The correlation between the ASQ scores and scenario failure or success (coded as 0 = failure and 1 = success) was −.40 ($N = 48, p < .01$). This result showed that participants who successfully completed a scenario tended to give lower (more favorable) ASQ ratings and provided evidence of concurrent validity.

Sensitivity. Of the 48 participants, 27 completed all of the ASQ items for all of the scenarios. This reduced database was appropriate for an analysis of variance (ANOVA; see Table 2) to assess the sensitivity of the ASQ. Specifically, did the ASQ scores discriminate among the different systems, user groups, or scenarios in the three usability studies?

The main effect of scenario was highly significant $F(7, 126) = 8.92, p < .0001$. The Scenario × System interaction was also significant, $F(14, 126) = 1.75, p = .05$. These results suggest that the ASQ scale score is a reasonably sensitive measure.

Table 2. ASQ Sensitivity Analysis

Source	*df*	Type III *SS*	*MS*	*F*	*p*
Total	215	6343.73	29.51		
System	2	29.81	14.91	0.15	.86
Group	2	458.51	229.26	2.29	.13
System × Group	4	34.82	8.71	0.09	.98
Error	18	1801.48	100.08		
Scenario	7	980.50	140.07	8.92	.0001
Scenario × System	14	383.44	27.39	1.74	.05
Scenario × Group	14	283.77	20.27	1.29	.22
Scenario × System × Group	28	393.07	14.04	0.89	.62
Error (Scenario)	126	1978.33	15.70		

Note. The total *df* for this analysis was 215 (27 participants times 8 scenario scores, minus 1 df for the grand mean), with *df* partitioned as shown above.

Discussion

The psychometric evaluation of this questionnaire showed that it is reasonable to condense the three ASQ items into a single scale through summation (or, equivalently, averaging). The available evidence indicates that the ASQ is reliable, valid, and sensitive. This condensation should allow easier interpretation and reporting of results when usability practitioners use the ASQ.

THE POST-STUDY SYSTEM USABILITY QUESTIONNAIRE (PSSUQ)

The Post-Study System Usability Questionnaire (PSSUQ) is currently a 19-item instrument for assessing user satisfaction with system usability (see Appendix for a copy of questionnaire items). Participants need about 10 min to complete the PSSUQ, but only complete it once, at the end of a usability study. Completing the PSSUQ allows participants to provide an overall evaluation of the system they used.

Development History of the PSSUQ

As part of a major software development project, a team of IBM usability experts constructed the items for the PSSUQ using a list of usability attributes known to influence user perception of usability (similar to the attributes described by Whiteside et al., 1988). The original version of the PSSUQ had only 18 items, with the items in a different order than shown in the Appendix (see Lewis, 1992b, for a copy of the original version of the PSSUQ). Recently, a series of investigations using decision support systems revealed a common set of five system characteristics associated with usability by several different user groups (Doug Antonelli, personal communication, January 5, 1991). Because the original 18-item PSSUQ addressed only four of these five system characteristics, I added an additional item (Item 8 in the current version) to address the fifth of these five system characteristics (improved productivity). However, the analyses reported in this section of this article used the original 18-item version of the questionnaire. In these analyses, the current Item 8 was not available (see the next section on the CSUQ for analyses that include Item 8).

Item Construction

The items are 7-point graphic scales, anchored at the end points with *strongly agree* (1) and *strongly disagree* (7) and a *not applicable* (N/A) point outside the scale.

Item Selection

A group of usability evaluators selected the items on the basis of their comprehensive content regarding hypothesized constituents of usability. For example, the items assess such system characteristics as ease of use, ease of learning, simplicity, effectiveness, information, and the user interface.

Psychometric Evaluation

The Office-Applications Studies. After the 48 participants in the office-applications usability study (Lewis et al., 1990) completed all the scenarios, they rated their system with the PSSUQ (see the section on the ASQ for more details about the study). This data allowed preliminary psychometric evaluation of the PSSUQ (Lewis, 1992b).

Factor Analysis. Discontinuity analysis (Coovert & McNelis, 1988) of the scree plot for an exploratory principal factors analysis of the PSSUQ data indicated that a three-factor solution was appropriate (see Figure 1), so the overall scale defined by the full set of items contained three subscales. Table 3 shows the varimax-rotated factor pattern, revealing the structure of the subscales. Boldface type in Table 3 highlights factor loadings that exceeded .50. Items that loaded highly on two factors were ambiguous regarding the appropriate subscale of which they should be a component, so they did not become a component of any subscale (see Appendix to examine the content of these items).

One of the most difficult tasks following this type of exploratory factor analysis is naming the factors. After considering a number of alternatives, a group of human factors engineers named the factors (and their corresponding subscales) System Usefulness (SYSUSE), Information Quality (INFOQUAL), and Interface Quality (INTERQUAL). These three factors account for 87% of the variability in the data.

Reliability. Coefficient alpha analyses showed that the reliability of the overall summative scale (OVERALL) was .97, and ranged from .91 to .96 for the three subscales (SYSUSE = .96, INFOQUAL = .91, and INTERQUAL = .91). Therefore, the overall scale and the three subscales have excellent reliability.

Validity. Correlation analyses support the validity of the scales. Using data from the 22 participants who completed all ASQ and all PSSUQ items, the OVERALL scale correlated highly with the sum of the ASQ ratings that participants gave after completing each scenario, $r(20) = .80, p = .0001$. Using data from the 31 participants who completed all scenarios and all items on the PSSUQ, OVERALL also correlated significantly with the percentage of successful scenario completion, $r(29) = -.40, p = .026$. Thirty-eight participants completed all SYSUSE items and all scenarios, and 37 participants completed all INTERQUAL items and all

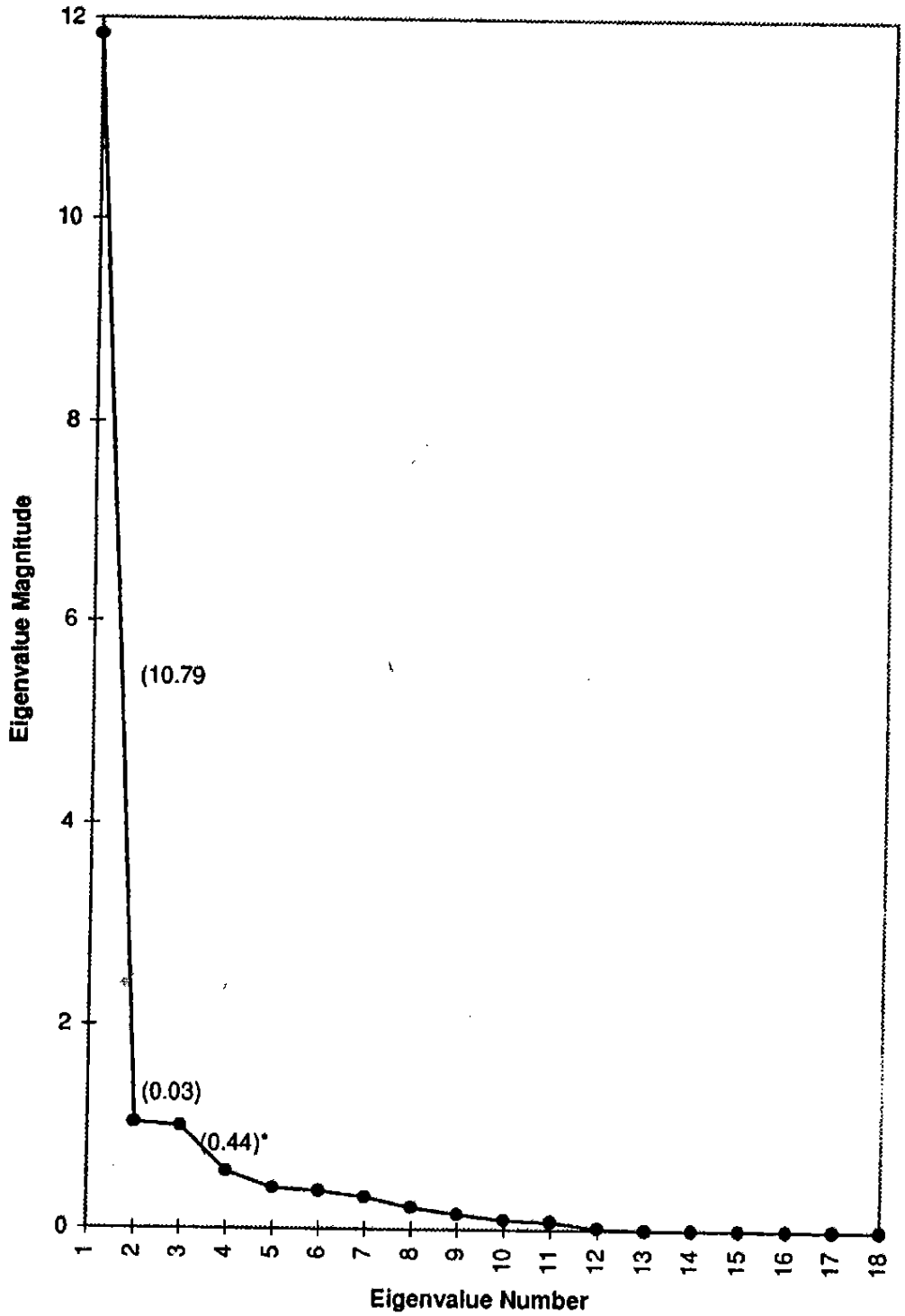


Figure 1. Scree plot (marking first discontinuity with an asterisk for the PSSUQ principal factors analysis; negative eigenvalues appear as 0.

Table 3. Varimax-Rotated Factor Pattern for the Principal Factor Analysis of the PSSUQ

Item	Subscale	Factor 1	Factor 2	Factor 3
1	1	.77	.26	.43
2	1	.63	.35	.46
3	1	.75	.38	.25
4	1	.81	.45	.07
5	1	.80	.16	.36
6	1	.68	.37	.48
7	1	.69	.46	.40
8[a]	N/A			
9	2	.05	.61	.24
10	2	.36	.71	.23
11	2	.45	.63	.25
12	2	.44	.75	.22
13	2	.43	.70	.32
14	2	.43	.74	.40
15	N/A	.30	.59	.56
16	3	.30	.36	.75
17	3	.37	.36	.76
18	3	.22	.28	.80
19	N/A	.58	.21	.64

[a]The first version of the PSSUQ (Lewis, 1992a) did not contain this item, so it does not contribute to this analysis.

scenarios. The SYSUSE, $r(36) = -.40, p = .006$, and INTERQUAL, $r(35) = -.29, p = .08$, scores correlated with the percentage of successful scenario completion.

Sensitivity. Table 4 shows the ANOVA tables for sensitivity analyses of the PSSUQ's OVERALL, SYSUSE, INFOQUAL, and INTERQUAL scales. In the sensitivity ANOVAs, the overall scale and all three subscales indicated significant differences among the user groups: OVERALL, $F(2, 29) = 4.35, p = .02$; SYSUSE, $F(2, 36) = 6.9, p = .003$; INFOQUAL, $F(2, 33) = 3.68, p = .04$; INTERQUAL, $F(2, 33) = 3.74, p$ .03. INFOQUAL showed a significant system effect, $F(2, 33) = 3.18, p = .05$).

In addition to the univariate ANOVAs, a multivariate analysis of variance (MANOVA) using SYSUSE, INFOQUAL, and INTERQUAL as dependent measures for generalized F tests (F_0; see Cliff, 1987) detected significant differences among systems, $F_0(2, 2, 28) = 11.53, p < .01$, and marginally significant differences among groups, $F_0(2, 2, 28) = 5.74, p < .10$, but no system by group interaction, $F_0(3, 2, 28) = 3.8, p > .20$.

Discussion

These findings have limited generalizability because the sample size for the factor analysis was relatively small. The usual recommendation would require 90 participants to factor analyze this questionnaire. However, the factor analysis and reliability analyses suggest that it is reasonable to define three subscales from

Table 4. PSSUQ Sensitivity Analyses

Source	*df*	Type III *SS*	*MS*	*F*	*p*
OVERALL					
Total	37	65.26	1.76		
System	2	1.13	0.57	0.38	.69
Group	2	12.98	6.49	4.35	.02
System × Group	4	10.79	2.70	1.81	.15
Error	29	43.26	1.49		
SYSUSE					
Total	44	94.03	1.76		
System	2	1.73	0.87	0.51	.60
Group	2	23.21	11.61	6.90	.003
System × Group	4	10.99	2.75	1.63	.19
Error	36	60.60	1.68		
INFOQUAL					
Total	41	71.55	1.75		
System	2	8.69	4.35	3.18	.05
Group	2	10.04	5.02	3.68	.04
System × Group	4	10.28	2.57	1.88	.14
Error	33	45.02	1.36		
INTERQUAL					
Total	43	84.18	1.76		
System	2	2.52	1.26	0.77	.47
Group	2	12.35	6.18	3.74	.03
System × Group	4	12.35	3.09	1.87	.14
Error	35	43.26	1.49		

Note. For the OVERALL analysis, 38 participants completed all PSSUQ items; 45 participants completed all SYSUSE items; 42 participants completed all INFOQUAL items; and 44 participants completed all INTERQUAL items.

this set of items. The PSSUQ has reasonable concurrent validity when compared with successful scenario completion rates and the ASQ scores. The overall scale and the subscales are reasonably sensitive. The evidence provided sufficient justification to use the PSSUQ to measure user satisfaction with system usability in usability studies, but also suggested that it would be prudent to collect more data in different circumstances to extend the generalizability of the findings.

THE COMPUTER SYSTEM USABILITY QUESTIONNAIRE (CSUQ)

The PSSUQ research was preliminary for two reasons. First, the sample size for the factor analysis was small, consisting of data from only 48 participants. Second, the PSSUQ data came from a usability study. This setting may have influ-

enced the correlations among the items and, therefore, the resultant factors. The purpose of this research (Lewis, 1992a) was to use a slightly revised version of the PSSUQ, the Computer System Usability Questionnaire (CSUQ) to obtain a database of sufficient size to calculate stable factors from a mailed survey. If the same factors emerged from this research as from the PSSUQ research, the study would demonstrate the potential usefulness of the questionnaire across different user groups and different research settings.

Item Selection and Construction

The CSUQ is identical to the PSSUQ (Lewis, 1991b), except that the wording of the items does not refer to a usability testing situation. For example, Item 3 of the PSSUQ states, "I could effectively complete the tasks and scenarios using this system," but Item 3 of the CSUQ states, "I can effectively complete my work using this system" (see Appendix for CSUQ items).

Psychometric Evaluation

The Mail Survey Using the CSUQ. The participants were 825 randomly selected IBM employees who worked at nine IBM development sites: Atlanta, Austin, Bethesda, Boca Raton, Dallas, Raleigh, Rochester, San Jose, and Tucson. Participants received a copy of the CSUQ with a cover letter asking them to use the CSUQ to rate their satisfaction with their primary computer system. Responses from questionnaires returned within 3 months of mailing made up the database for this study.

Factor Analysis. Forty-six percent (n = 377) of the participants returned the questionnaire. A principal factor analysis of the returned questionnaires produced the scree plot shown in Figure 2. The scree plot was similar to that found for the PSSUQ, and discontinuity analysis indicated that an appropriate factor analysis should solve for three factors. (Figure 2 also shows the scree plot and discontinuity analysis for a maximum-likelihood factors analysis of the data, also indicating a three-factor solution.)

Table 5 shows the varimax-rotated three-factor solution for the principal factor analysis. The selection criterion for the factor loadings was .50, indicated in boldface type in the table. The factor analysis showed that Item 8 ("I believe I became productive quickly using this system"), which was not part of the original PSSUQ, should be part of Factor 1. Item 15 ("The organization of information on the system screens is clear"), which loaded on two factors in the PSSUQ study, loaded only on Factor 2 in the current study. In the PSSUQ study and in the current study, Item 19 ("Overall, I am satisfied with this system") loaded on both Factors 1 and 3, and is not part of any subscale. Otherwise, the factor structure of the CSUQ is very similar to that of the PSSUQ, so the CSUQ subscales have the

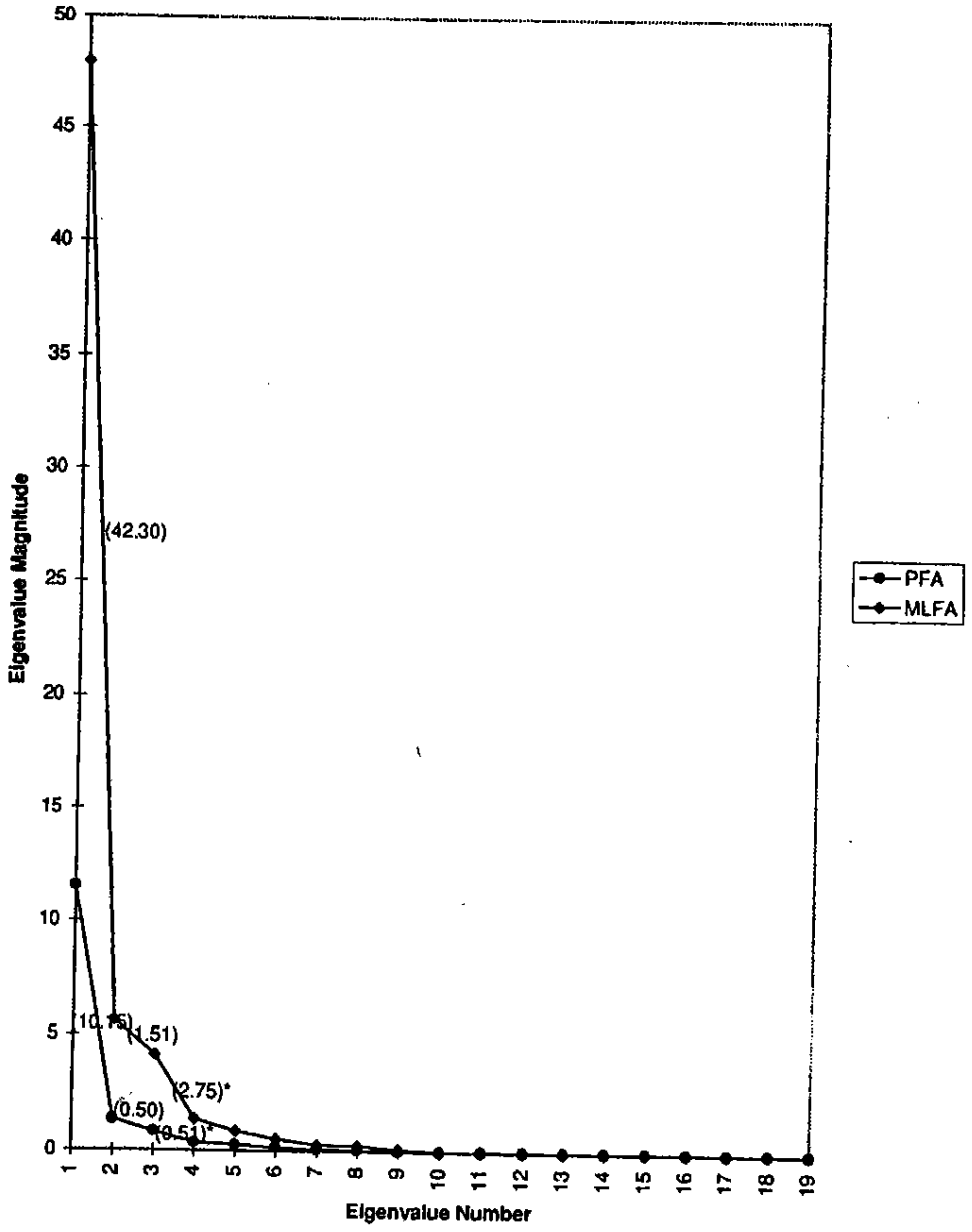


Figure 2. Scree plots (marking first discontinuity with an asterisk) for the CSUQ principal factors and maximum likelihood analyses; negative eigenvalues appear as 0.

same names as those of the PSSUQ. The three factors accounted for 98.6% of the variability in the rating data.

Reliability. In all cases, coefficient alpha exceeded .89, indicating acceptable scale reliability. The estimates of coefficient alpha for the CSUQ were .93 for SYSUSE, .91 for INFOQUAL, .89 for INTERQUAL, and .95 for the OVERALL scale. The values of coefficient alpha for the CSUQ scales were within .03 of those for the PSSUQ scales.

Table 5. Varimax-Rotated Factor Pattern for the Principal Factor Analysis of the CSUQ

Item	Subscale	Factor 1	Factor 2	Factor 3
1	1	.74	.36	.26
2	1	.69	.41	.16
3	1	.72	.21	.36
4	1	.74	.31	.33
5	1	.77	.30	.32
6	1	.72	.22	.27
7	1	.63	.49	.13
8	1	.66	.39	.26
9	2	.23	.72	.21
10	2	.34	.67	.28
11	2	.23	.81	.20
12	2	.24	.77	.27
13	2	.38	.76	.17
14	2	.40	.73	.18
15	2	.34	.57	.40
16	3	.33	.27	.81
17	3	.38	.26	.81
18	3	.34	.35	.56
19	N/A	.66	.37	.50

Validity/Sensitivity. After establishing scale reliability, the next step in psychometric evaluation is to determine scale validity. However, without a concurrent or predicted measurement, it is impossible to obtain a quantitative measure of validity in the traditional psychometric sense. An indirect way to assess validity is to examine scale sensitivity to variables that should systematically affect the scale. The sensitivity analyses of the PSSUQ (Lewis, 1992b) showed significant effects of user group (business professional with mouse experience, business professional without mouse experience, and secretary/clerk without mouse experience) on the OVERALL, SYSUSE, INFOQUAL, and INTERQUAL scales. The type of computer system the participant used during the study significantly affected the INFOQUAL scale.

A comprehensive listing of the influence of respondent characteristics on the CSUQ scores is outside the scope of this article. However, the significant findings are similar to those for the PSSUQ. The type of computer that respondents used significantly affected their responses only for the INFOQUAL score, $F(5, 311) = 2.14$, $p = .06$. The number of years of experience with their computer system affected respondents' scores for OVERALL, $F(4, 294) = 3.12$, $p = .02$, SYSUSE, $F(4, 332) = 2.05$, $p = .09$, INFOQUAL, $F(4, 311) = 2.59$, $p = .04$, and INTERQUAL, $F(4, 322) = 2.47$, $p = .04$. The respondents' range of experience with computer systems (number of different computer systems that they reported having used) affected scores for OVERALL, $F(3, 294) = 2.77$, $p = .04$, INFOQUAL, $F(3, 311) = 2.60$, $p = .05$, and INTERQUAL, $F(3, 322) = 2.14$, $p = .10$. These significant findings provide indirect support to the hypothesis that these scales are valid. (As with previous analyses, the error degrees of freedom vary according to the number of participants who completed all the items for the given scale.)

Discussion

The key results from this study are (a) a demonstration of stable factors for the CSUQ (and, by extension, for the PSSUQ), and (b) evidence that the questionnaire works well in nonlaboratory settings. The CSUQ scales are comparable to the PSSUQ scales, both in terms of reliability and validity (indicated by similarity in the sensitivity analyses). These findings substantially enhance the usefulness of the CSUQ and PSSUQ to usability practitioners. Researchers who conduct usability studies (either laboratory or nonlaboratory) can use this questionnaire to assess user satisfaction with system usability.

GENERAL DISCUSSION

Although user satisfaction with system usability is only one component of the multifaceted construct of usability (Bevan et al., 1991), it is a very important component in many situations. It is especially important when a primary design goal is user satisfaction. This article has described the psychometric qualities of IBM questionnaires that assess user satisfaction with system usability: the ASQ, PSSUQ, and CSUQ.

The ASQ is an after-scenario questionnaire, intended for use in a scenario-based usability testing situation. Its psychometric evaluation showed acceptable estimates of reliability, validity, and sensitivity. The PSSUQ and CSUQ are both overall satisfaction questionnaires. The PSSUQ items are appropriate for a usability testing situation, and the CSUQ items are appropriate for a field-testing situation. Otherwise, the questionnaires are identical. The psychometric evaluations of the PSSUQ (using data from a usability study) and the CSUQ (using data from a mail survey) were virtually identical. This consistency provides strong evidence of generalizability of results and wide applicability of the questionnaires.

Because these questionnaires have acceptable psychometric properties, usability practitioners can use them with confidence as standardized measurements of satisfaction for usability studies and tests (ASQ, PSSUQ) or field research (CSUQ). (Practitioners should note that nothing prevents the addition of items to these questionnaires if a particular situation suggests the need. However, using these questionnaires as the foundation for special-purpose questionnaires ensures that practitioners can score the scales and subscales from the questionnaires, maintaining the advantages of standardized measurement.) In conclusion, these questionnaires should be valuable additions to the repertoire of techniques that usability practitioners can apply in the evaluation of usable systems.

One limitation of the current research with these questionnaires is that all the validity measures were concurrent. However, the availability of these questionnaires should allow future research to address questions of predictive validity. With three similar (but not identical) measuring instruments now available (CSUQ, QUIS, and CUSI), an ambitious researcher could attempt to construct a

multitrait–multimethod matrix to investigate convergent and discriminant validity for the construct of usability.

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APPENDIX. THE IBM QUESTIONNAIRES

The After-Scenario Questionnaire (ASQ)

Administration and Scoring. Give the questionnaire to a participant after he or she has completed a scenario during a usability evaluation. Average (with the arithmetic mean) the scores from the three items to obtain the ASQ score for a participant's satisfaction with the system for a given scenario. Low scores are better than high scores due to the anchors used in the 7-point scales. If a participant does not answer an item or marks N/A, average the remaining items to obtain the ASQ score.

Instructions and Items. The questionnaire's instructions and items follow. For each of the statements below, circle the rating of your choice.

1. Overall, I am satisfied with the ease of completing the tasks in this scenario.

strongly agree <——————————> strongly disagree

1 2 3 4 5 6 7 N/A

Comments:

2. Overall, I am satisfied with the amount of time it took to complete the tasks in this scenario.

strongly agree <——————————> strongly disagree

1 2 3 4 5 6 7 N/A

Comments:

3. Overall, I am satisfied with the support information (on-line help, messages, documentation) when completing the tasks.

strongly agree <——————————> strongly disagree

1 2 3 4 5 6 7 N/A

Comments:

The Post-Study System Usability Questionnaire (PSSUQ)

Administration and Scoring. Give the PSSUQ to participants after they have completed all the scenarios in a usability study. You can calculate four scores from the responses to the PSSUQ items: the overall satisfaction score (OVERALL),

system usefulness (SYSUSE), information quality (INFOQUAL), and interface quality (INTERQUAL). Because research on an alternative form of the PSSUQ (the Computer System Usability Questionnaire, or CSUQ) confirmed and clarified (and slightly modified) the factor structure of the questionnaire, refer to Table A1 in the next section for the current scoring rules of the PSSUQ.

Instructions and Items.

The PSSUQ items use the same format as that shown for the ASQ. The questionnaire's instructions and items follow.

This questionnaire gives you an opportunity to tell us your reactions to the system you used. Your responses will help us understand what aspects of the system you are particularly concerned about and the aspects that satisfy you. To as great a degree as possible, think about all the tasks that you have done with the system while you answer these questions. Please read each statement and indicate how strongly you agree or disagree with the statement by circling a number on the scale. If a statement does not apply to you, circle N/A. Please write comments to elaborate on your answers. After you have completed this questionnaire, I'll go over your answers with you to make sure I understand all of your responses. Thank you!

1. Overall, I am satisfied with how easy it is to use this system.
2. It was simple to use this system.
3. I could effectively complete the tasks and scenarios using this system.
4. I was able to complete the tasks and scenarios quickly using this system.
5. I was able to efficiently complete the tasks and scenarios using this system.
6. I felt comfortable using this system.
7. It was easy to learn to use this system.
8. I believe I could become productive quickly using this system.
9. The system gave error messages that clearly told me how to fix problems.
10. Whenever I made a mistake using the system, I could recover easily and quickly.
11. The information (such as on-line help, on-screen messages, and other documentation) provided with this system was clear.
12. It was easy to find the information I needed.
13. The information provided for the system was easy to understand.
14. The information was effective in helping me complete the tasks and scenarios.
15. The organization of information on the system screens was clear.

Note. The interface includes those items that you use to interact with the system. For example, some components of the interface are the keyboard, the mouse, and the screens (including their use of graphics and language).

16. The interface of this system was pleasant.
17. I liked using the interface of this system.

18. This system has all the functions and capabilities I expect it to have.
19. Overall, I am satisfied with this system.

The Computer System Usability Questionnaire (CSUQ)

Administration and Scoring. Use the CSUQ rather than the PSSUQ when the usability study is in a nonlaboratory setting. Table A1 contains the rules for calculating the CSUQ and PSSUQ scores. Average the scores from the appropriate items to obtain the scale and subscale scores. Low scores are better than high scores due to the anchors used in the 7-point scales. If a participant does not answer an item or marks N/A, then average the remaining item scores.

Table A1. Rules for Calculating CSUQ/PSSUQ Scores

Score Name	Average the Responses to
OVERALL	Items 1–19
SYSUSE	Items 1–8
INFOQUAL	Items 9–15
INTERQUAL	Items 16–18

Instructions and Items. The CSUQ items use the same format as that shown for the ASQ. The questionnaire's instructions and items follow.

This questionnaire gives you an opportunity to express your satisfaction with the usability of your primary computer system. Your responses will help us understand what aspects of the system you are particularly concerned about and the aspects that satisfy you. To as great a degree as possible, think about all the tasks that you have done with the system while you answer these questions. Please read each statement and indicate how strongly you agree or disagree with the statement by circling a number on the scale. If a statement does not apply to you, circle N/A. Whenever it is appropriate, please write comments to explain your answers. Thank you!

1. Overall, I am satisfied with how easy it is to use this system.
2. It is simple to use this system.
3. I can effectively complete my work using this system.
4. I am able to complete my work quickly using this system.
5. I am able to efficiently complete my work using this system.
6. I feel comfortable using this system.
7. It was easy to learn to use this system.
8. I believe I became productive quickly using this system.
9. The system gives error messages that clearly tell me how to fix problems.
10. Whenever I make a mistake using the system, I recover easily and quickly.

11. The information (such as on-line help, on-screen messages, and other documentation) provided with this system is clear.
12. It is easy to find the information I need.
13. The information provided with the system is easy to understand.
14. The information is effective in helping me complete my work.
15. The organization of information on the system screens is clear.

Note. The interface includes those items that you use to interact with the system. For example, some components of the interface are the keyboard, the mouse, the screens (including their use of graphics and language).

16. The interface of this system is pleasant.
17. I like using the interface of this system.
18. This system has all the functions and capabilities I expect it to have.
19. Overall, I am satisfied with this system.